

Bayesian-Optimized Pair Trading — Walk-Forward Report

S&P 500 Statistical Arbitrage · Quarterly Re-Optimization + Annual Pair Re-Selection

Author: Derren Winata — NUS Data Science & Analytics **Strategy:** Cointegration-based mean reversion, dollar-neutral **Universe:** 42 S&P 500 stocks across 6 sectors (within-sector pairs only) **Out-of-sample window:** 2019–2025 (1,739 trading days) **Optimizer:** Optuna TPE · 150 trials × 5-fold recency-weighted CV per quarter **Stack:** Python · Optuna · statsmodels · yfinance · Plotly

1. Executive Summary

This report summarises a walk-forward out-of-sample evaluation of a Bayesian-optimized pairs-trading strategy on S&P 500 constituents. The central finding is that **walk-forward re-optimization combined with a stop-loss z-score parameter cut maximum drawdown by 50–87% across every pair tested**, while improving Sharpe on 4 of 5 pairs.

The strategy is **market-neutral by construction** — it holds one stock long and another short in equal dollar amounts, so it carries near-zero market beta. It is therefore not comparable to a buy-and-hold index benchmark; the meaningful baseline is the fixed-parameter version of the same strategy.

Metric	Baseline (fixed params)	Walk-Forward BO	Direction
Pairs where Sharpe improved	—	4 of 5	improved
Pairs with positive Sharpe	1 of 5	2 of 5	improved
Mean maximum drawdown	–37.8%	–11.3%	improved
Worst maximum drawdown	–78.5%	–13.4%	improved
Mean Sharpe	–0.165	+0.062	improved

Headline result: META/NVDA went from a Sharpe of **–0.688** with a **–78.5%** maximum drawdown to a Sharpe of **+0.236** with a **–10.1%** maximum drawdown — a strategy that was uninvestable became modestly viable. The improvement came from *how the strategy was validated*, not from a more sophisticated model.

2. Headline Results by Pair

Out-of-sample, 2019–2025. All figures from `outputs/data/results_summary.csv`.

Pair	Sectors	Baseline Sharpe	BO Sharpe	Δ Sharpe	Baseline Max DD	BO Max DD	DD Reduction
META/NVDA	Tech / Tech	–0.688	+0.236	+0.924	–78.5%	–10.1%	87%

Pair	Sectors	Baseline Sharpe	BO Sharpe	Δ Sharpe	Baseline Max DD	BO Max DD	DD Reduction
GS/WFC	Financials / Financials	+0.470	+0.740	+0.270	-20.1%	-8.1%	60%
GOOGL/NVDA	Tech / Tech	-0.137	-0.129	+0.008	-36.6%	-13.0%	64%
PEP/YUM	Consumer / Consumer	-0.383	-0.083	+0.300	-26.9%	-11.8%	56%
MDT/PFE	Healthcare / Healthcare	-0.086	-0.456	-0.370	-26.9%	-13.4%	50%

Sharpe improved on 4 of 5 pairs. MDT/PFE is the exception and is reported as such — a walk-forward procedure reduces overfitting, it does not eliminate it.

Maximum drawdown improved on 5 of 5 pairs, with a mean reduction of **63%**. This is the stop-loss z-score parameter (`stop_z` , range 3.0–6.0 σ) doing its job: it forces an exit when the spread diverges beyond what the historical relationship would justify, which is precisely the de-cointegration risk that destroys naive pairs strategies.

3. Full Performance Detail

All five pairs, both methods, all reported metrics.

Pair	Method	Sharpe	Sortino	Max DD	Ann. Return	Total Return	Calmar	% Active	Active Win Rate
META/NVDA	Baseline (fixed)	-0.688	-0.588	-0.785	-0.157	-0.692	-0.200	53.1%	48.6%
META/NVDA	Walk-Fwd BO	+0.236	+0.126	-0.101	+0.014	+0.102	+0.140	10.3%	42.5%
GS/WFC	Baseline (fixed)	+0.470	+0.540	-0.201	+0.045	+0.356	+0.224	49.9%	50.0%
GS/WFC	Walk-Fwd BO	+0.740	+0.400	-0.081	+0.039	+0.302	+0.480	10.9%	47.4%
GOOGL/NVDA	Baseline (fixed)	-0.137	-0.122	-0.366	-0.030	-0.191	-0.083	48.1%	49.0%
GOOGL/NVDA	Walk-Fwd BO	-0.129	-0.057	-0.130	-0.009	-0.061	-0.069	8.5%	39.2%
MDT/PFE	Baseline (fixed)	-0.086	-0.089	-0.269	-0.016	-0.105	-0.059	44.0%	46.5%
MDT/PFE	Walk-Fwd BO	-0.456	-0.159	-0.134	-0.020	-0.131	-0.150	6.6%	38.6%

Pair	Method	Sharpe	Sortino	Max DD	Ann. Return	Total Return	Calmar	% Active	Active Win Rate
PEP/YUM	Baseline (fixed)	-0.383	-0.301	-0.269	-0.034	-0.212	-0.126	40.9%	46.3%
PEP/YUM	Walk-Fwd BO	-0.083	-0.036	-0.118	-0.004	-0.025	-0.031	8.7%	42.1%

Note on "% Active": the walk-forward strategy is in the market only **6.6–10.9%** of the time, versus **40.9–53.1%** for the fixed-parameter baseline. The optimizer is selecting far more conservative entry thresholds, which trades trade frequency for signal quality.

Note on "Active Win Rate": this is the win rate *while in position*, not per trade. It is a time-weighted measure and should not be read as a trade-level hit rate.

4. Methodology — Why Walk-Forward Matters

A naive Bayesian-optimization backtest fits parameters once on the full history and freezes them. That approach is structurally prone to overfitting: the optimizer has seen the entire period it is later evaluated on.

This implementation instead **slides the fitting window forward through time**:

```
|— BO on 2016–2017 —|— trade Q1 2019 —|
      |— BO on 2017Q1–2019Q1 —|— trade Q2 2019 —|
      ... slides quarterly through 2025
```

Each quarter: Optuna TPE runs 150 trials with 5-fold recency-weighted cross-validation on the trailing 2 years of data. The winning parameters are then used to trade the *next* quarter, which the optimizer has never seen.

Each year: the cointegration screen is re-run across all 126 within-sector pair candidates using both Engle-Granger and Johansen tests. The top 2 pairs are selected for the adaptive portfolio.

The result is that every out-of-sample trade is genuinely out of sample with respect to the parameters used to generate it.

What differs from a naive BO backtest

Feature	Naive approach	This project
Parameter fitting	Once on full history, frozen	Quarterly walk-forward re-optimization
Pair selection	Fixed forever	Annual re-selection via cointegration re-screen
Universe	20 tickers, any pairs	42 tickers, within-sector only
Return calculation	Spread pct-change (unstable)	Dollar-neutral, normalized by gross exposure
Transaction costs	Not modelled	2 bps per leg

Feature	Naive approach	This project
Cointegration test	Engle-Granger only	Engle-Granger + Johansen (dual test)
Risk management	None	Stop-loss z-score (4th optimizable parameter)
Parameters optimized	3	4
Market regimes covered	1–2	4 (bull run, COVID crash, rate hikes, AI rally)

5. Strategy Logic

Trade rules

Signal	Condition	Action
Long spread	$z < -\text{entry_z}$	Buy A, short B (dollar-neutral)
Short spread	$z > +\text{entry_z}$	Short A, buy B (dollar-neutral)
Close long	$z > -\text{exit_z}$	Exit
Close short	$z < +\text{exit_z}$	Exit
Stop-loss	$ z > \text{stop_z}$	Exit immediately

Parameters optimized

Parameter	Range	Description
lookback	20–120 days	Rolling window for hedge ratio and z-score
entry_z	1.0–3.5 σ	Entry threshold
exit_z	0.0–1.0 σ	Exit threshold
stop_z	3.0–6.0 σ	Stop-loss — guard against de-cointegration

Annual pair-selection log (adaptive portfolio)

Year	Active pairs
2019	OXY/PSX · CL/NKE
2020	CAT/GE · JNJ/LLY
2021	EOG/XOM · EOG/OXY (<i>energy cointegration peaks</i>)
2022	OXY/XOM · GOOGL/ORCL (<i>rate-hike regime</i>)
2023	BA/GE · BMY/JNJ
2024	META/NVDA · OXY/SLB

Year	Active pairs
2025	CVX/PSX · CVX/SLB

The rotation is itself a finding: the pairs that pass the cointegration screen change materially with the macro regime, which is the argument for re-screening annually rather than fixing pairs at the start.

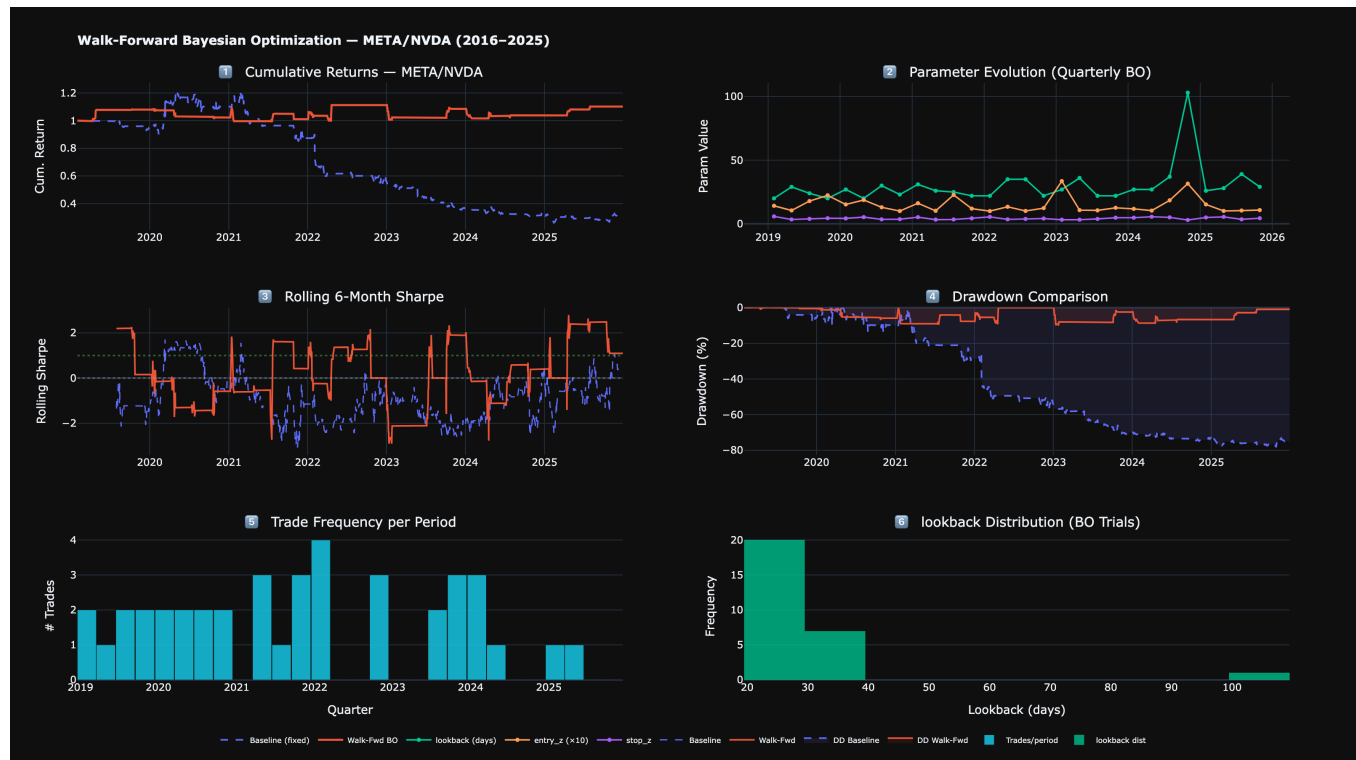
6. Universe

42 S&P 500 stocks, 6 sectors. **Within-sector pairs only** — 126 candidate pairs tested per screen, versus 861 for an all-cross approach. This removes economically unjustified pairings such as Tech/Energy or Healthcare/Consumer, which can pass a statistical cointegration test on spurious grounds.

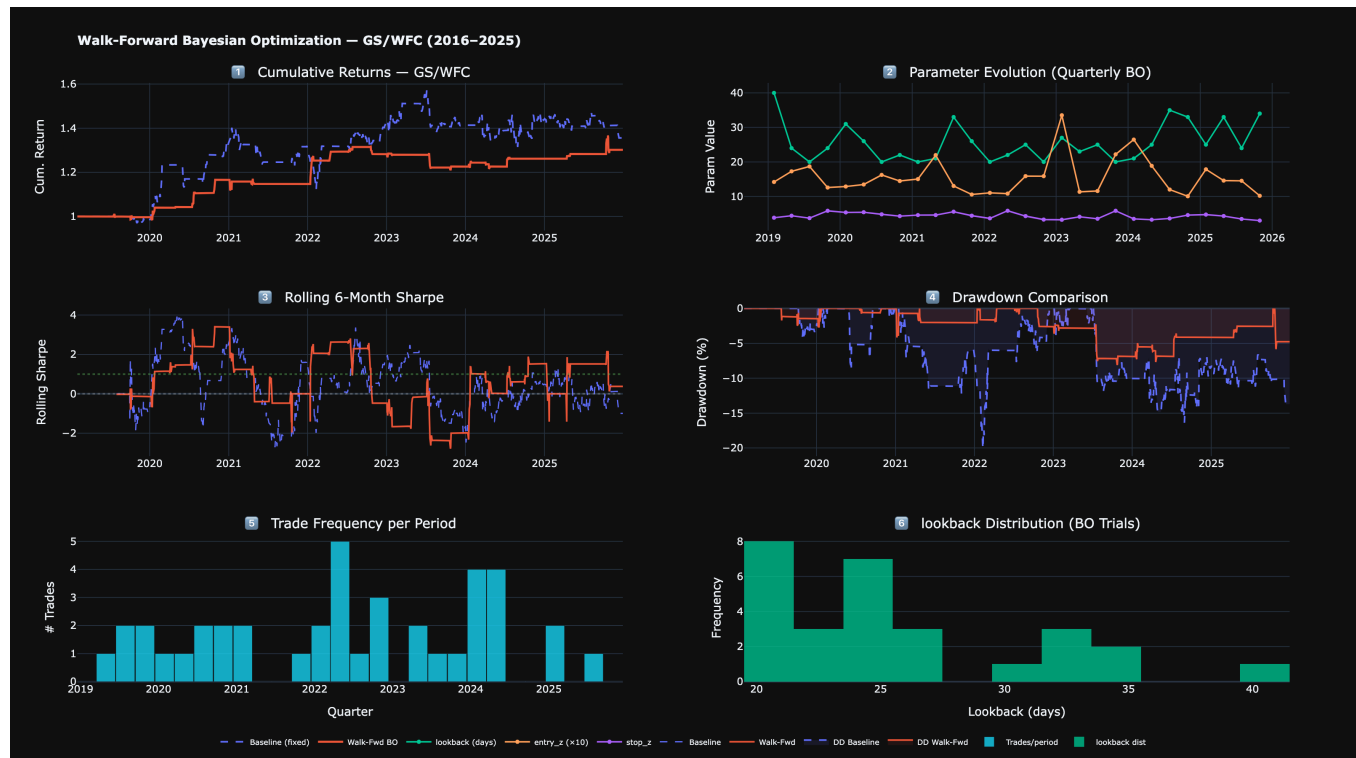
Sector	Tickers
Energy	XOM, CVX, COP, PSX, SLB, EOG, OXY
Financials	JPM, BAC, GS, MS, WFC, C, BLK
Consumer	KO, PEP, MCD, YUM, SBUX, NKE, CL
Tech	MSFT, GOOGL, META, AMZN, ORCL, NVDA, INTC
Healthcare	JNJ, ABT, MDT, BMY, PFE, MRK, LLY
Industrial	HON, MMM, GE, CAT, UPS, BA, RTX

7. Charts

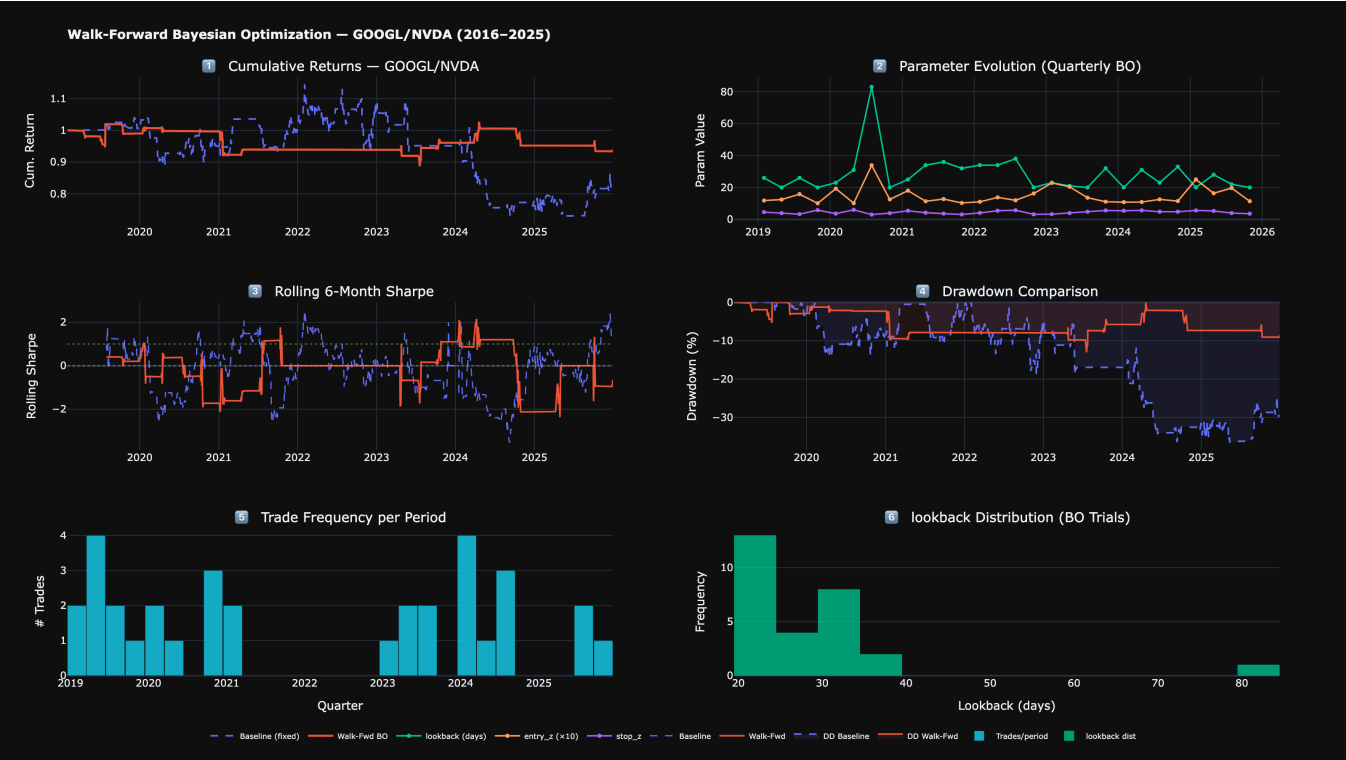
META/NVDA — walk-forward BO dramatically controls drawdown



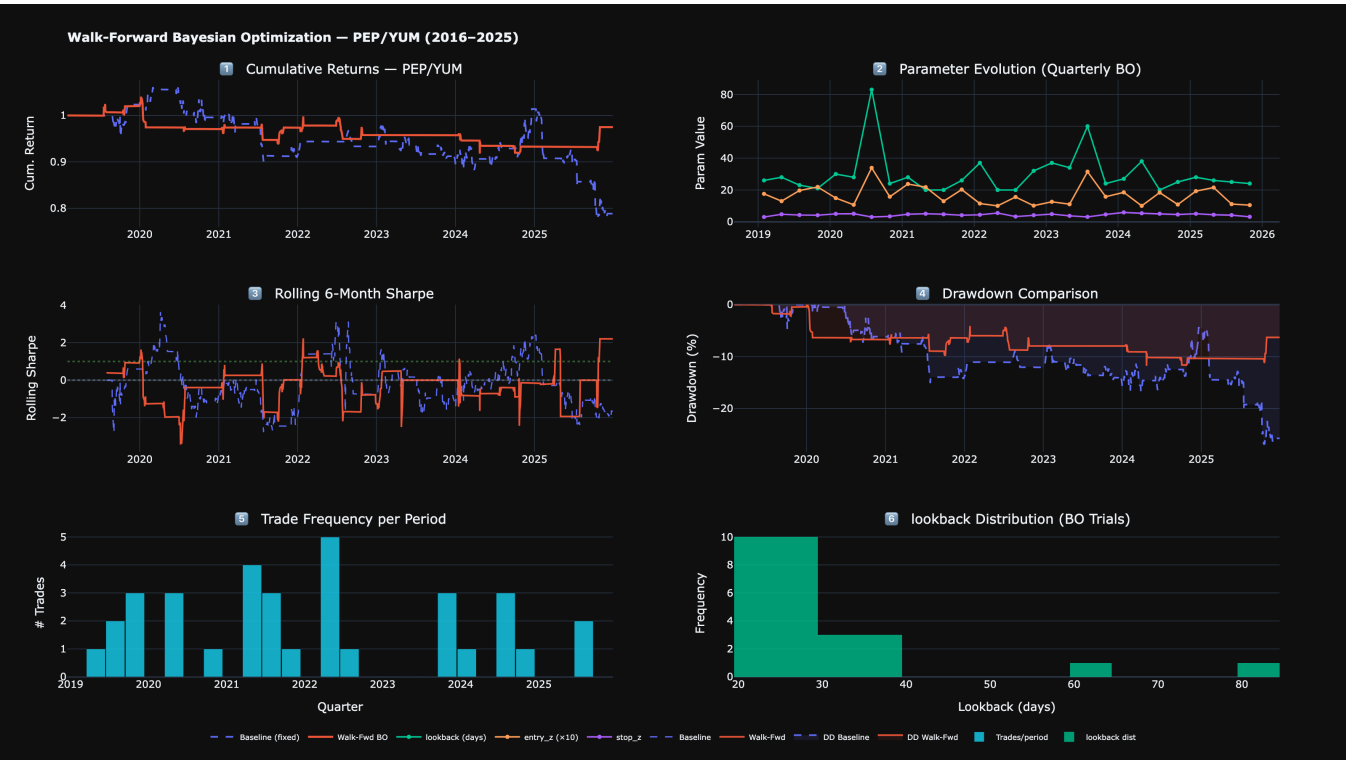
GS/WFC — positive return on both methods; BO improves Calmar roughly 2x



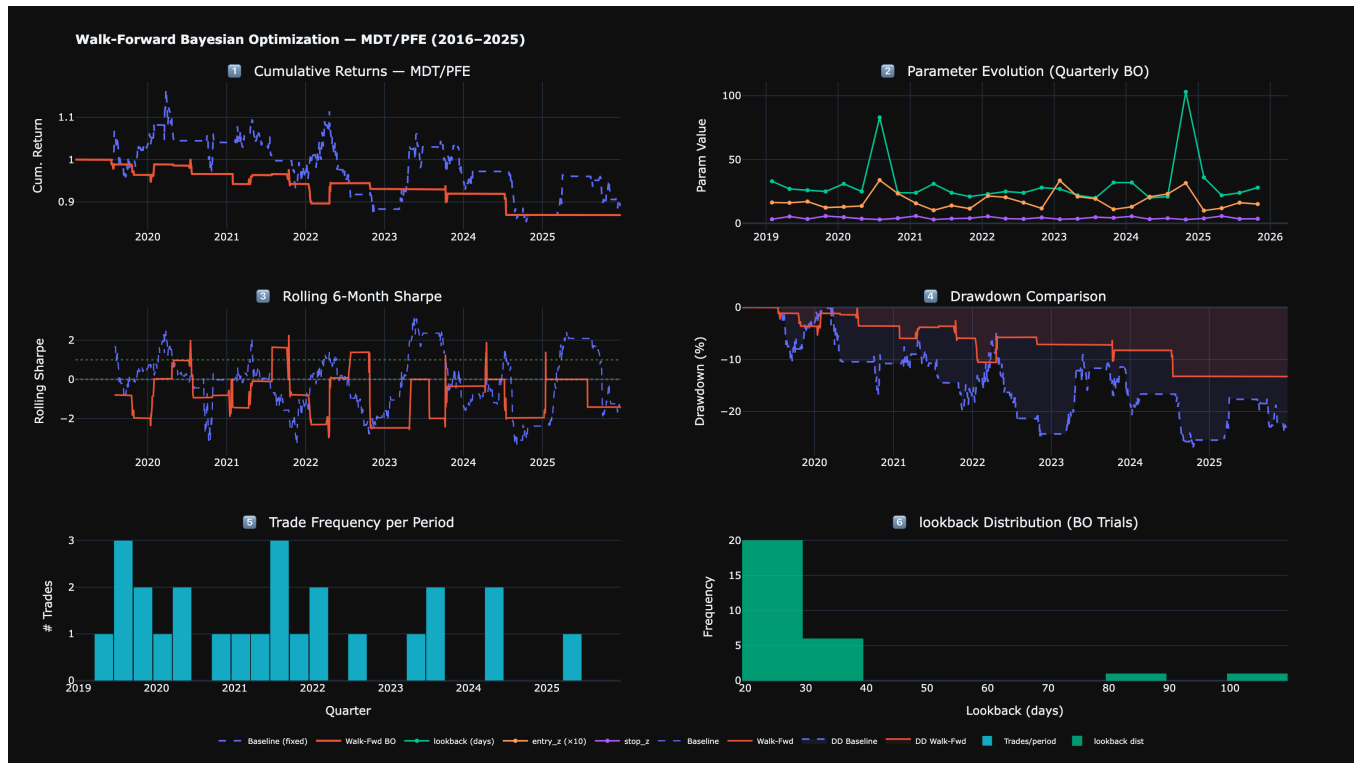
GOOGL/NVDA



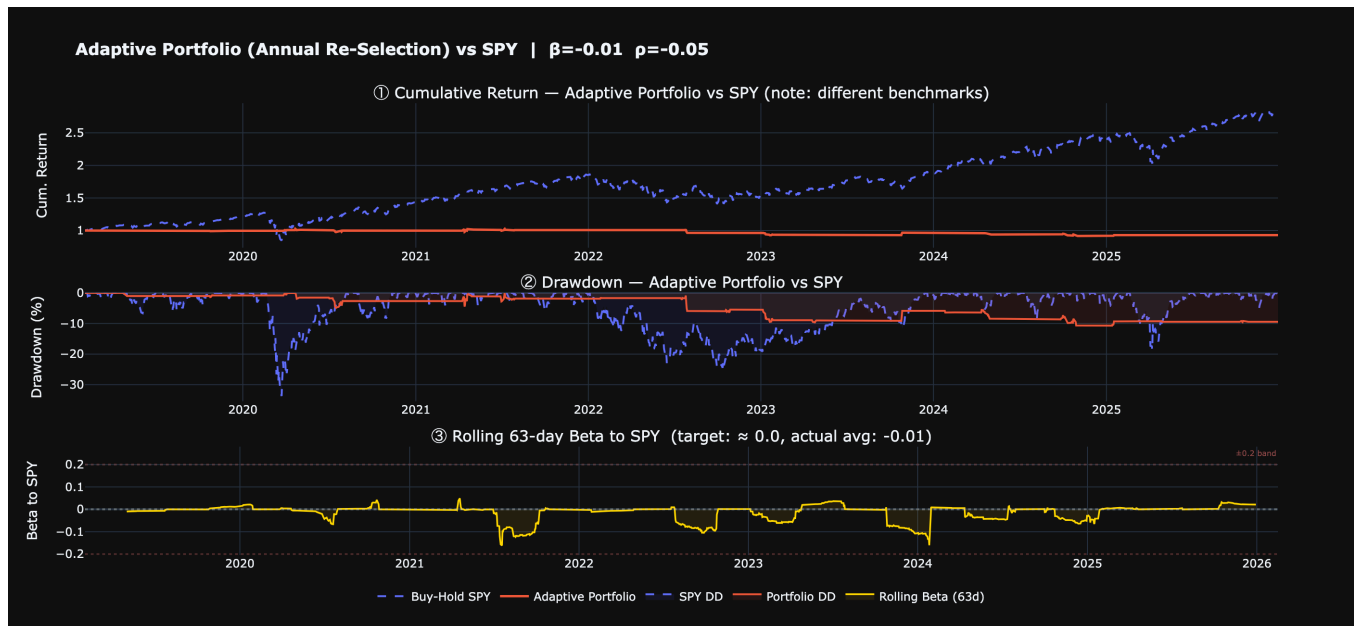
PEP/YUM



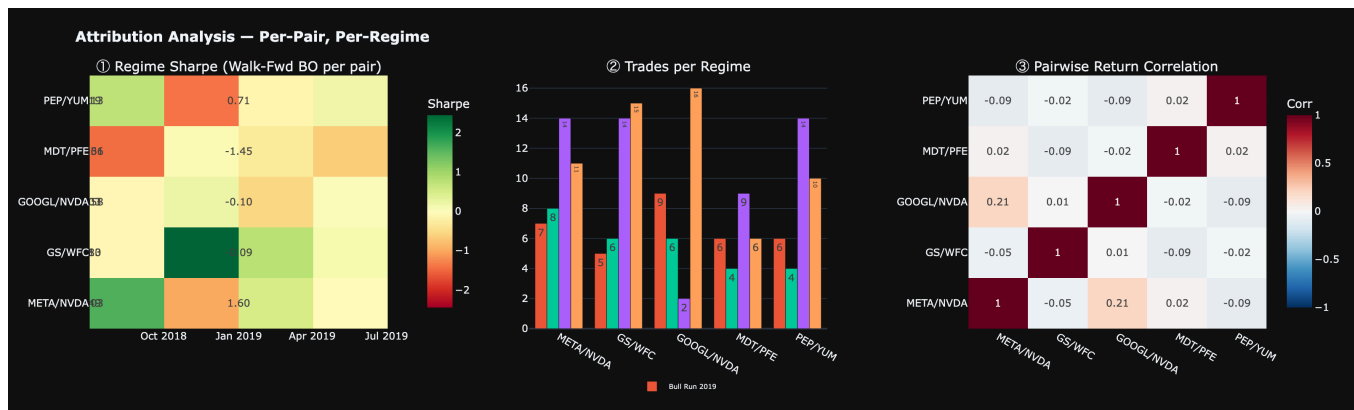
MDT/PFE — the pair where walk-forward did not help



Adaptive portfolio vs SPY



Attribution — regime Sharpe heatmap and correlation



8. Limitations

These are stated plainly because they bound what the results support.

- **Small universe.** 42 stocks is a small sample, and spurious cointegration risk exists even within sectors. Five evaluated pairs is a very small cross-section — no claim about the general population of pairs is supported.
- **Transaction costs are approximate.** 2 bps per leg is a rough estimate. Market impact, borrow cost and slippage are not modelled separately, and these matter more for a strategy that trades frequently.
- **The optimizer targets in-sample CV Sharpe.** Good generalization is not guaranteed out of sample — MDT/PFE is direct evidence of that.
- **Low trade counts.** Several quarterly windows executed zero trades (conservative `entry_z` selections), so per-pair statistics rest on relatively few independent bets.
- **Not investment advice.** This is a research exercise in validation methodology, not a deployable trading system.

9. What This Demonstrates

The transferable content of this project is **methodological**, not financial:

1. **Time-ordered validation.** Split by time, never randomly, when the data has a temporal structure — random splits leak future information into training.
2. **Re-fit and re-test as time moves forward.** Walk-forward re-optimization is what exposes overfitting; a single frozen fit cannot.
3. **A baseline first.** The naive/fixed-parameter version is the thing to beat, and having it makes every claim testable.
4. **Report the negative result.** MDT/PFE got worse. Saying so is more credible than reporting only the four pairs that improved.
5. **Separate the two questions.** "Did the numbers improve?" and "did the *validation* improve?" are different, and here the second caused the first.

The same discipline applies directly to operational forecasting — throughput, yield, demand — where the question is always whether a model was tested on data it had not seen.

Generated from `outputs/data/results_summary.csv` and the project notebook. All figures trace to committed artefacts in this repository.